

Muhammad Murtaza

AI Agent Systems Architect | Senior AI Agent Developer | Multi-Agent Pipeline Engineer | LLM Orchestration Specialist
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PROFESSIONAL SUMMARY

Senior Python AI Engineer with 5+ years of production experience designing and deploying AI agent systems, LLM-powered applications, and RAG pipelines that automate real business workflows at scale. Proven track record building multi-agent architectures with LangChain, LangGraph, and LlamaIndex; fine-tuning LLMs (Llama 3, GPT-4) with LoRA/QLoRA; and designing RESTful microservice APIs (FastAPI, Flask) serving 100K+ daily requests at sub-200ms latency. Delivered systems handling 50K+ monthly users and 2M+ document pipelines. Reduced inference latency by 25% and cloud costs by 30% through quantization and infrastructure optimization. Proficient with DeepEval and Ragas for LLM evaluation, Docker/Kubernetes for containerization, and AWS/Azure/GCP for cloud deployment. Strong communicator comfortable collaborating across engineering, product, and data science teams.

TECHNICAL SKILLS

Python & Frameworks: Python (5+ yrs), FastAPI, Flask, Django, Node.js, NestJS

AI Agents & Orchestration: LangChain, LangGraph, LangSmith, LlamaIndex, Multi-Agent Orchestration, Tool Integration, Stateful Memory, Human-in-the-Loop

LLMs & Generative AI: GPT-4/GPT-5, Claude (Anthropic), Llama 3, Gemini, Mistral, AWS Bedrock, Groq, Azure OpenAI

LLM Fine-Tuning: LoRA, QLoRA, Unsloth, 4-bit Quantization, Hugging Face Transformers, PEFT

RAG & Retrieval: RAG Pipeline Design, Semantic Search, Hybrid Search, Pinecone, FAISS, ChromaDB, Weaviate, Milvus

LLM Evaluation: DeepEval, Ragas, Prompt Engineering, Structured Output (JSON mode, Function Calling)

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, Text Classification, NER, Summarization, Sentiment Analysis

Prototyping Tools: Streamlit, Gradio

Speech & Voice AI: Whisper ASR, ElevenLabs TTS, Deepgram, Voice Pipelines

Frontend: React, Next.js 15, TypeScript, TailwindCSS, Framer Motion, ReactFlow

Cloud & MLOps: AWS (SageMaker, Lambda, EC2, Bedrock), Azure (OpenAI, ML), GCP, Docker, Kubernetes, MLflow, CI/CD, Git

Databases & Infra: PostgreSQL, MongoDB, Redis, Elasticsearch, Neon, Convex, ImageKit

Security & APIs: REST APIs, Microservices, JWT Auth, HIPAA Compliance, OAuth, API Rate Limiting

Other Tools: Streamlit, Remotion, Inngest, Clerk, Twilio, Stripe, Git

EXPERIENCE

Generative AI Engineer

Dec 2025 - May 2026

Highbro Technology - Remote

- Built advanced multi-agent systems and voice integration pipelines for enterprise clients, leveraging LangChain, LangGraph, and LlamaIndex for complex workflow orchestration.
- Developed NLP solutions and machine learning architectures, including custom RAG pipelines and fine-tuned LLMs, to automate document processing and conversational AI tasks.
- Collaborated with cross-functional teams to deliver production-ready AI systems, ensuring scalability and performance optimization on cloud infrastructure.

Generative AI Engineer & Python Backend Developer

Jan 2023 - Aug 2025

Vision Byte Technologies - Islamabad, Pakistan

- Designed and deployed production multi-agent AI systems using LangChain, LangGraph, LangSmith, and LlamaIndex - automating complex business workflows with parallel agent orchestration and tool integration, achieving 40% efficiency improvement.
- Built production RAG pipelines handling 2M+ documents at 92% retrieval accuracy using Pinecone, FAISS, ChromaDB, Weaviate, and Milvus with hybrid semantic search strategies.
- Fine-tuned LLMs (Llama 3, GPT-4, domain-specific models) using LoRA, QLoRA, and Hugging Face Transformers; achieved 35% performance improvement over baseline through custom training strategies and 4-bit quantization.

- Developed scalable FastAPI/Flask RESTful microservice APIs serving 100K+ daily requests at sub-200ms latency; designed secure endpoints with JWT auth and API rate limiting.
- Integrated and orchestrated cloud-hosted foundation models via AWS Bedrock, Azure OpenAI, Groq, and Anthropic Claude; reduced inference cost by 30% through model quantization and efficient batching.
- Built LLM evaluation pipelines using DeepEval and Ragas to monitor retrieval quality, answer faithfulness, and hallucination rates across deployed RAG systems.

ML & Deep Learning Engineer

Jan 2021 - Nov 2022

Dot Coder - KPK, Pakistan

- Built and deployed machine learning and NLP systems in production using Python, TensorFlow, and PyTorch for text classification and sentiment analysis, achieving 89% accuracy on real-world datasets.
- Developed end-to-end ML pipelines covering data collection, preprocessing, model training, evaluation, and deployment with automated monitoring and alerting.
- Created scalable Flask REST APIs exposing ML models to frontend applications, handling reliable JSON-based communication across services.
- Built rapid prototypes with Streamlit dashboards to visualize model outputs and communicate results to stakeholders.
- Optimized model deployment using Docker containerization, ensuring consistent cross-platform performance and simplified CI/CD integration.
- Applied security best practices for model APIs including input validation, authentication, and protection against prompt injection patterns in NLP systems.

KEY PROJECTS

AI Travel Chatbot

LangGraph, FastAPI, Next.js, AWS Bedrock, Groq, Llama 3.1 70B, TailwindCSS 4, Framer Motion, Clerk, Convex, Redis, Docker, PostgreSQL
LangGraph, FastAPI, Next.js, AWS Bedrock, Groq, Llama 3.1 70B, TailwindCSS 4, Framer Motion, Clerk, Convex, Redis, Docker, PostgreSQL Built an AI-powered travel planning chatbot with 9+ specialized parallel agents orchestrated via LangGraph: flight resolution, hotel search, activity planning, budget analysis, visa/weather advisory, and itinerary generation. Implemented Human-in-the-Loop Gen-UI with interactive cards inside the chat interface - planning engine fires only after full context is captured via structured JSON outputs.

Healthcare Conversational AI Platform

LLM Fine-Tuning, NER, FastAPI, RAG, DeepEval

LLM Fine-Tuning, NER, FastAPI, RAG, DeepEval Built end-to-end conversational AI using fine-tuned LLMs, NER, and intent classification for patient engagement - processing 50K+ monthly interactions with 4.6/5 satisfaction rating. Integrated RAG pipeline over healthcare knowledge base with 92% retrieval accuracy using Ragas evaluation; implemented HIPAA-compliant data handling with encryption and audit logs.

Voice Medical Assistant - MediSynapse

Whisper ASR, RAG, Vision, TTS, HIPAA

Whisper ASR, RAG, Vision, TTS, HIPAA Developed HIPAA-compliant voice AI assistant combining RAG, computer vision, and speech recognition with custom TTS for hands-free medical workflow automation. Reduced manual clinical tasks by 45% through intelligent voice-driven interaction and document retrieval with real-time streaming responses.

EDUCATION

Bachelor of Science in Information Technology

Oct 2021 – Jun 2025

Kohat University of Science & Technology

Machine Learning, Natural Language Processing, Deep Learning, Data Science, Healthcare Informatics

CERTIFICATIONS

Certified Generative AI Engineer & AI Agent Developer

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